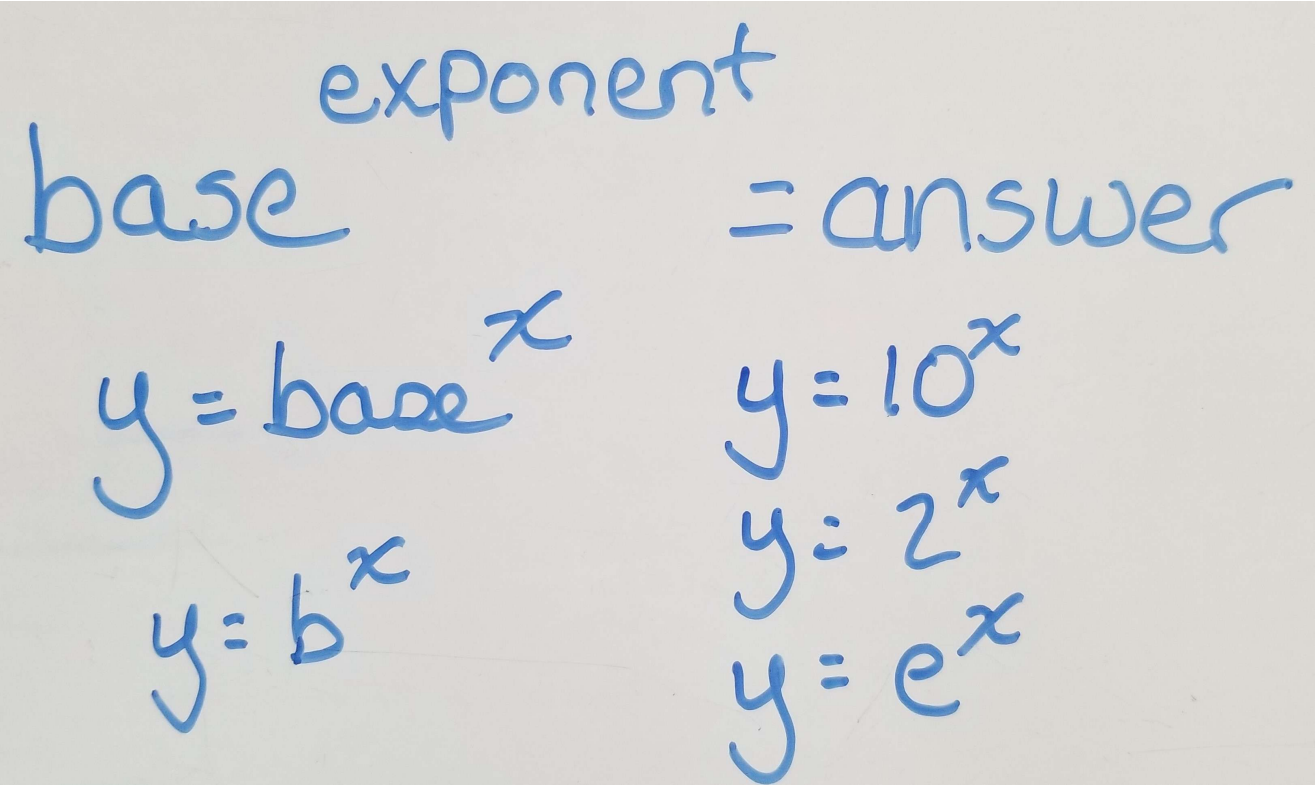
AG1

Two ways of thinking about exponential functions:

A base to an exponent equals an answer is useful for doing algebra.

y or f(x) equals e to the x is useful as a function or for calculus.



## Slide 1

**AG1** Reviewed

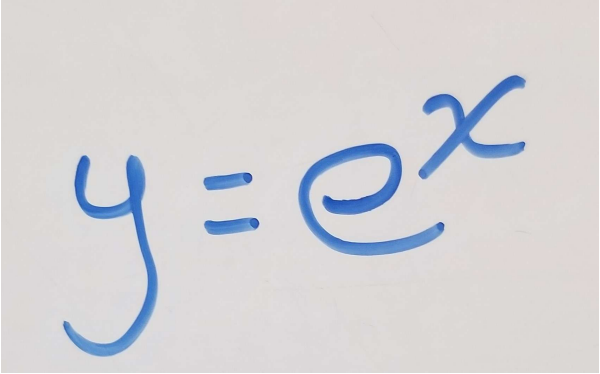
Alan Geller, 2026-01-24T22:10:10.157

# Base e or Euler's number

e (mathematical constant) - Wikipedia

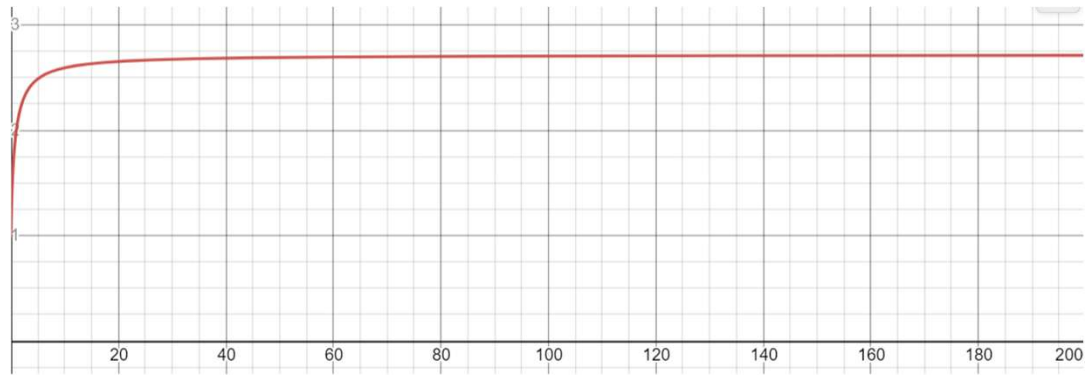
The standard exponential function is written e to the x. Where e is an irrational number that is about

2.718281828459045235360287471352662497757247093699 95 according to Wikipedia.



There are a number of different ways of finding, defining, and calculating e. The top is used for finance and interest rates. The bottom uses series notation, and the bottom is n! which we will learn when we do series and then probability.

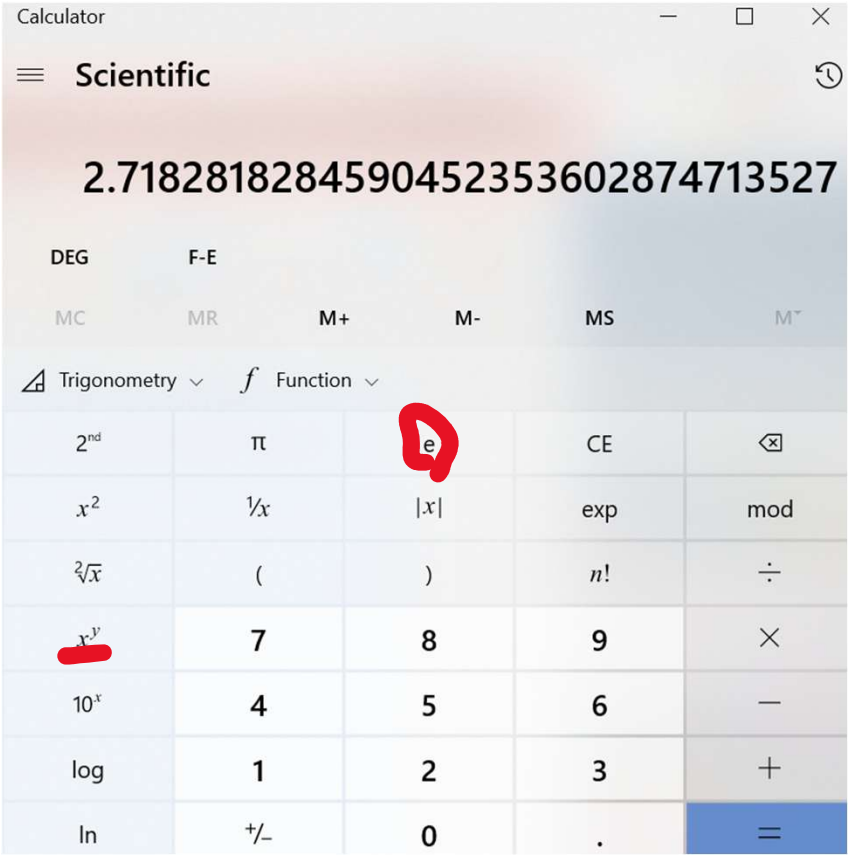
$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n$$




$$e \approx \frac{1}{1} + \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{1 \cdot 2 \cdot 3 \cdot 4} + \cdots$$

Approximate powers of e fall between 2^x and 3^x and are closer to 3^x and you can find e on scientific calculators.

| | |
|---|---|
| e | 2.718 |
| e^2 | 7.389 |
| e^3 | 20.085 |
| e^4 | 54.598 |



The exponential function $\boldsymbol{e^x}$ is the only function that is its own derivative.

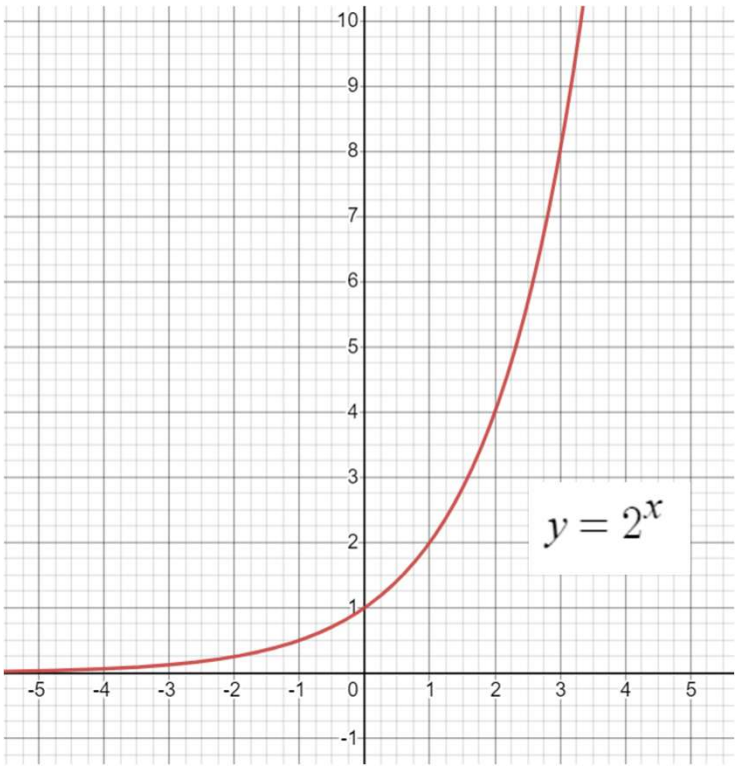
The number *e* such that the graph of y= $\boldsymbol{e^x}$ that the graph of y= $\boldsymbol{e^x}$ has a slope of 1 when x=0.

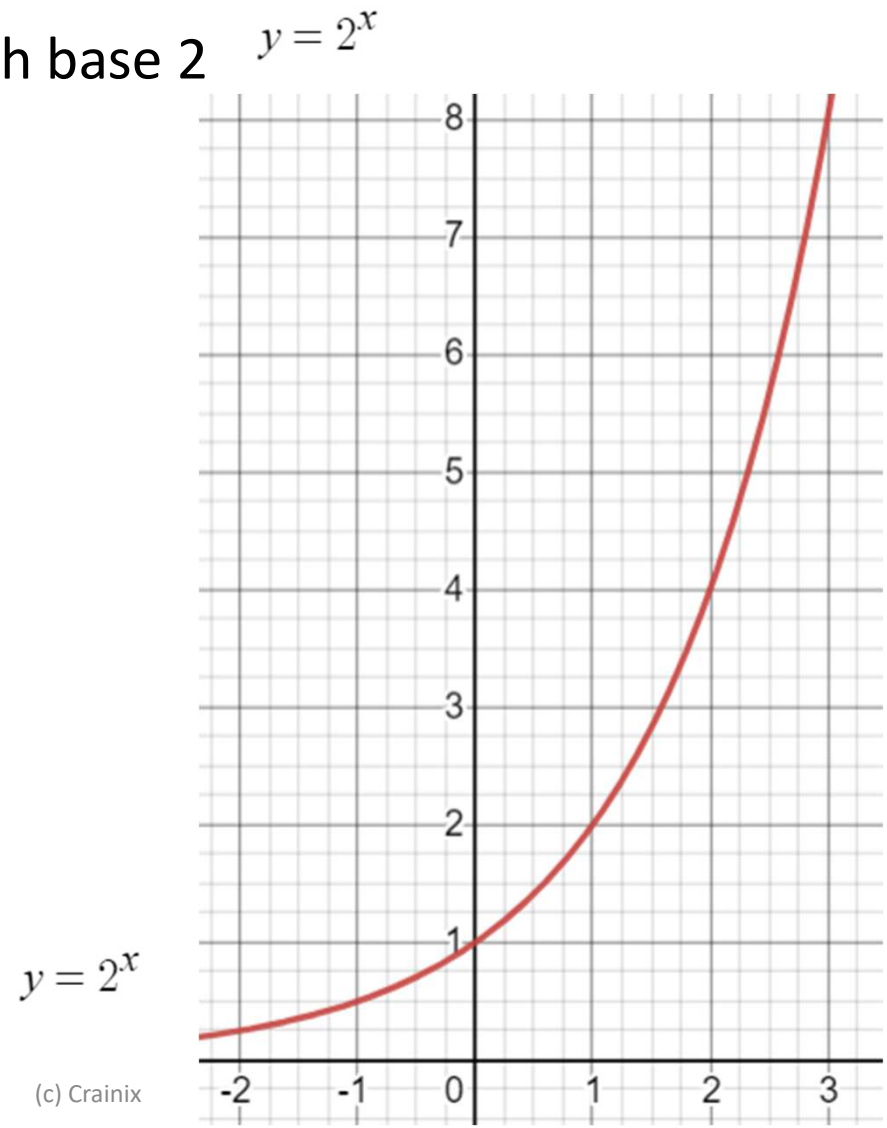
$$y = e^x$$

$$y'(0) = 1$$

It is easier to see this from a graph.

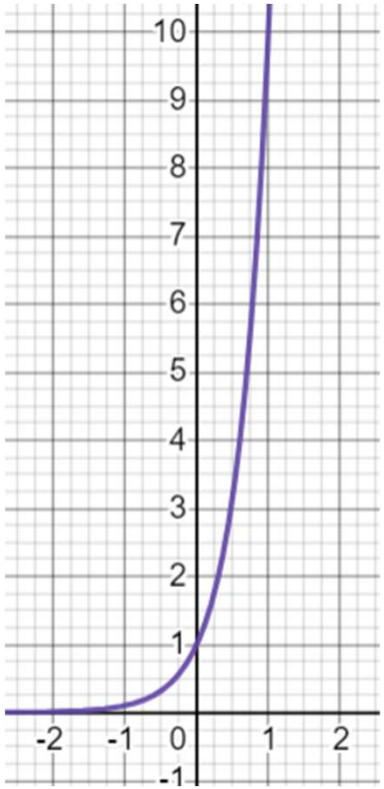
# Graphing exponential functions with base 2

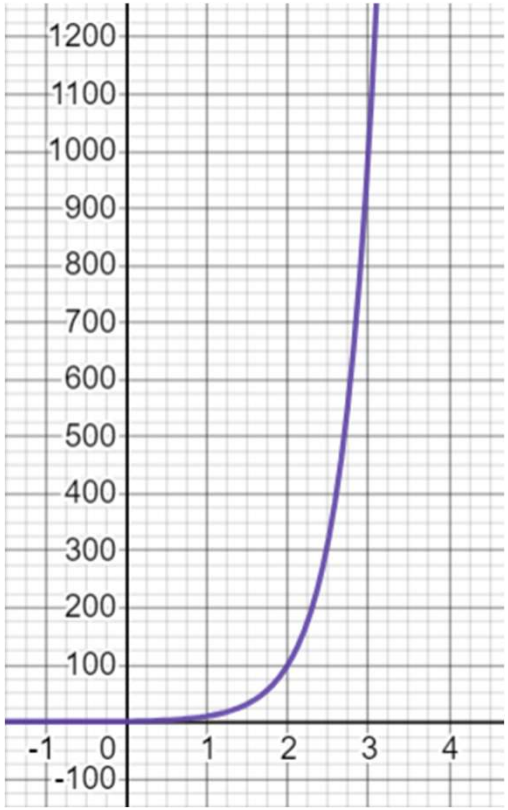


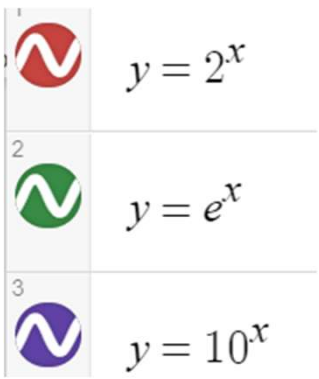
# Graphing exponential functions with base 10 $y = 10^x$

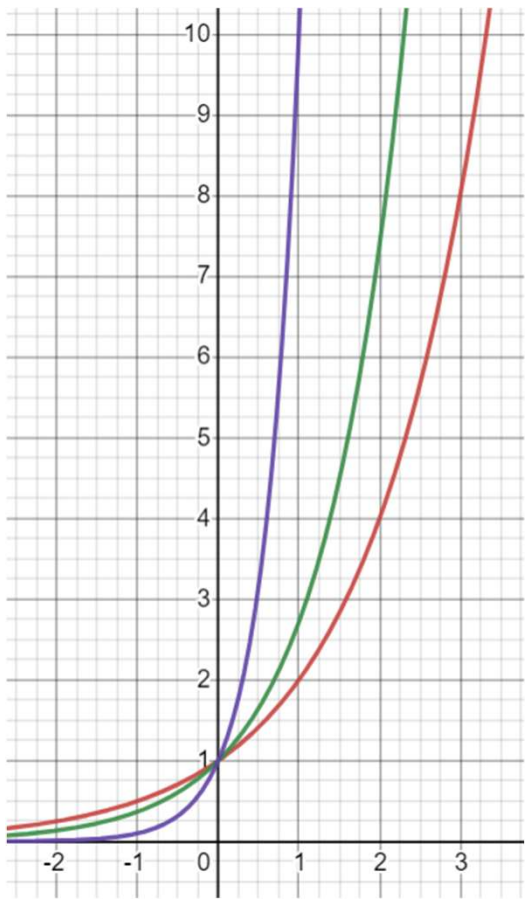
$y = 10^x$

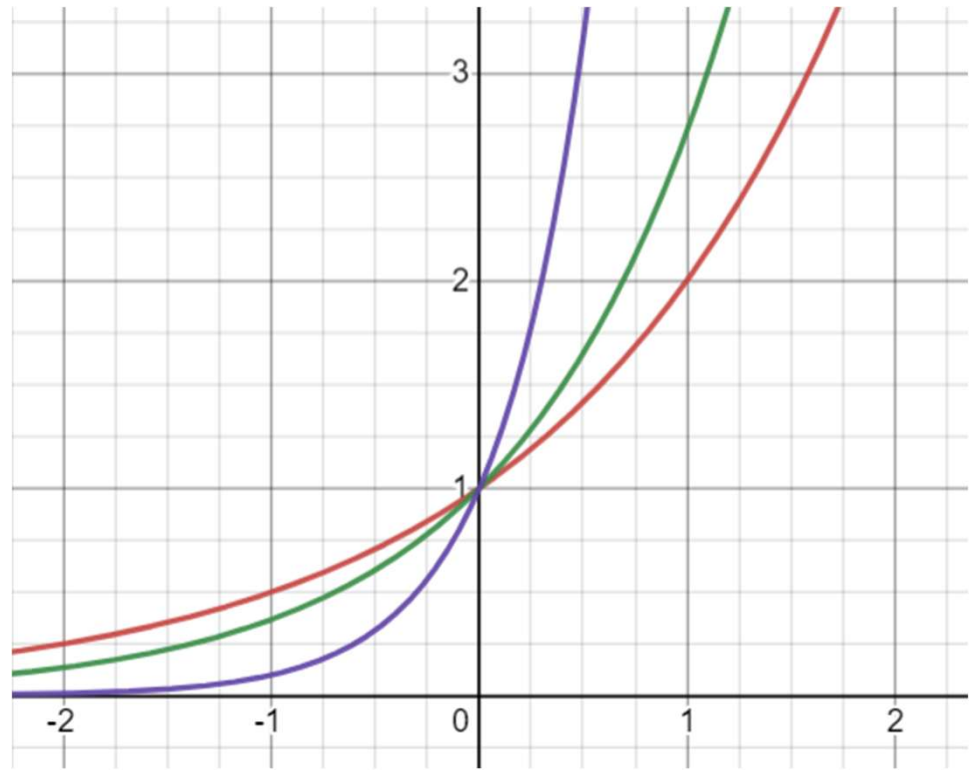


Graphing exponential functions with base 2, e, and 10.



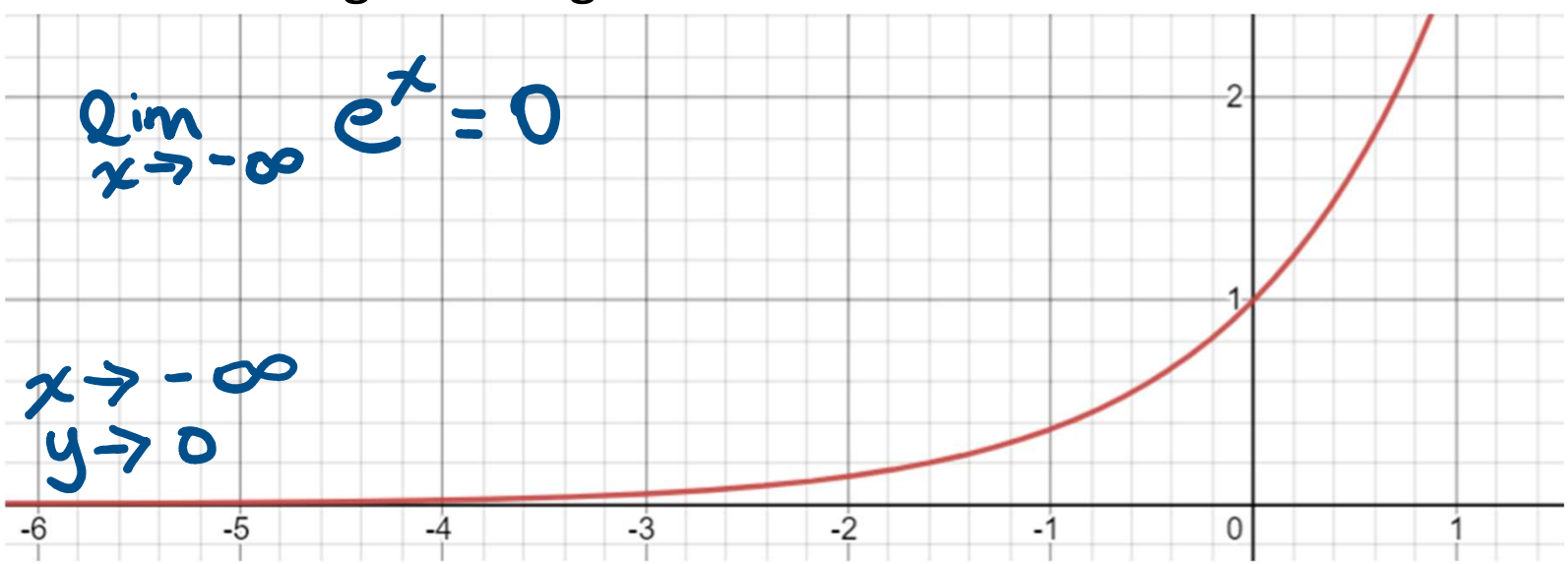



Exponential function has a horizontal asymptote at y=0.

An **asymptote** is a line that the function approaches but can never cross. It is a value that the function gets close to but can never equal. We can talk about the limit of the function as x approaches negative infinity as being zero. (Asymptote is usually used for vertical.)

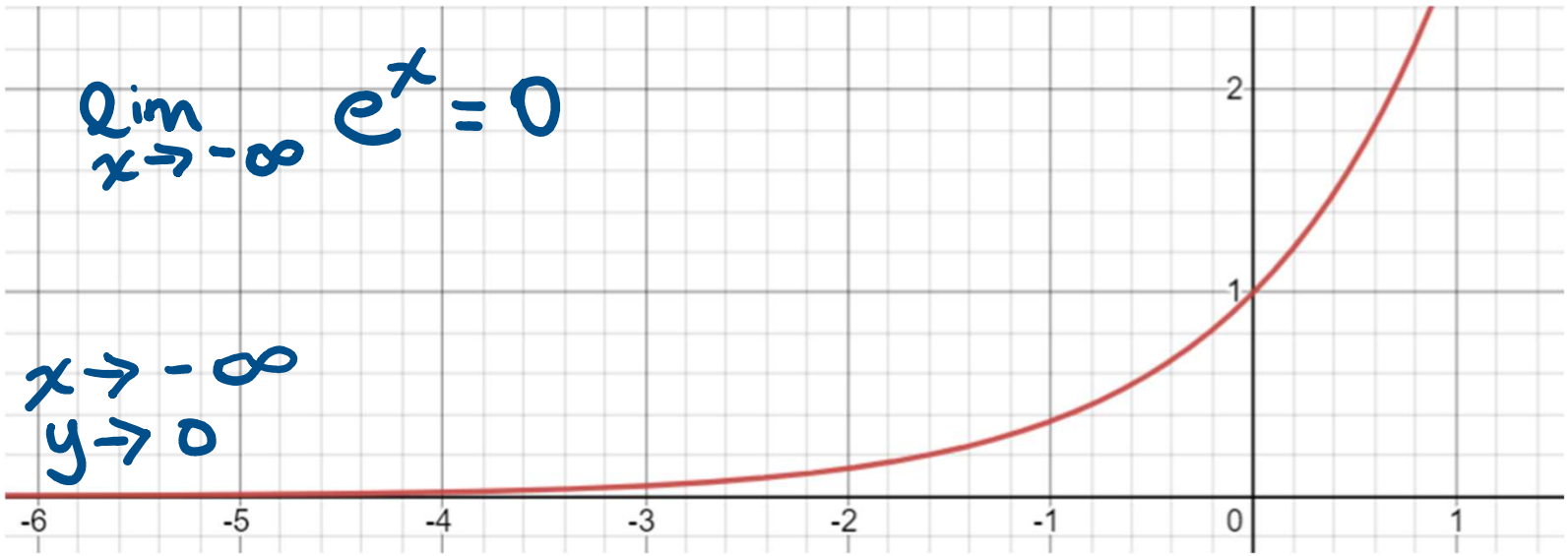
You can think of a **limit** or a function asymptotically going towards a wall but never being able to get there.



You can think of a **limit** or an asymptote as going towards a wall but never being able to get there.

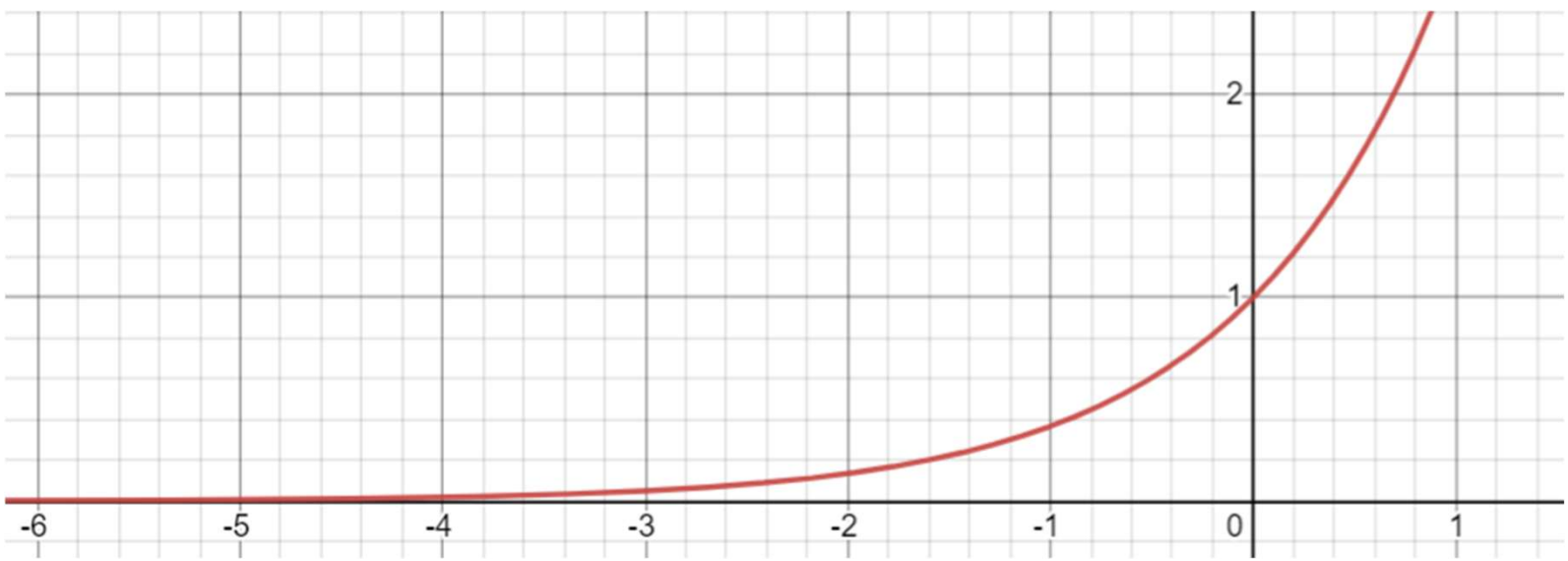
For 2 to the x, you can think of it as going halfway towards a wall, getting smaller and smaller as you go so you never actually hit the wall.

The wall is the limit and we say the limit equals the wall.



We can talk about the limit of the function as x approaches negative infinity as being zero. Here is how to write that algebraically:

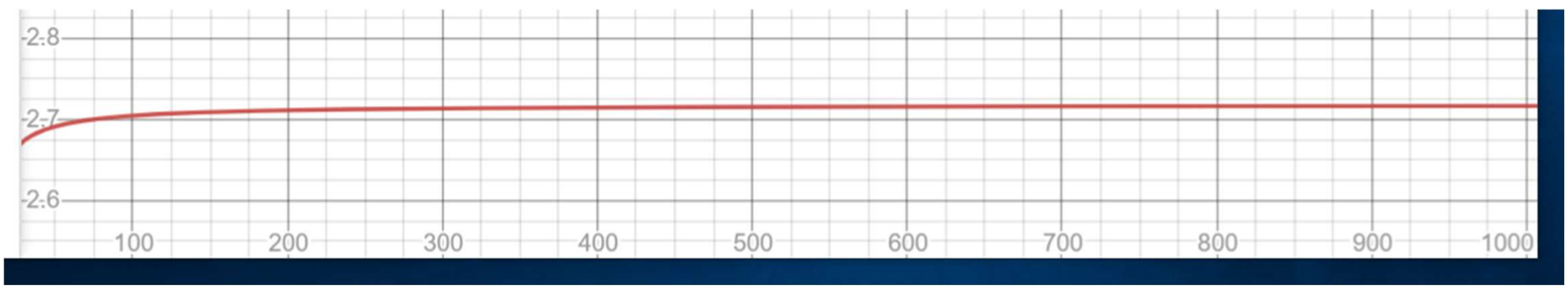
$$\lim_{x \to -\infty} e^x = 0$$



The number e can be approximated by the equation below as x gets infinitely larger. It is the limit that the function approaches or the number that the output gets closer to as the input, or x gets bigger. As x gets bigger, 1/x gets infinitely small, but it is multiplied by itself almost infinitely many times. The number e is irrational and gets close to 2.718… e (mathematical constant) – Wikipedia

$$y = \left(1 + \frac{1}{(x)}\right)^x$$

2.718281828459045235360287471352 7



Showing the tangent line, which is the derivative, to the exponential function at e^0=1

$f(x) = e^x$

2

$g(x) = \frac{d}{dx} f(x)$

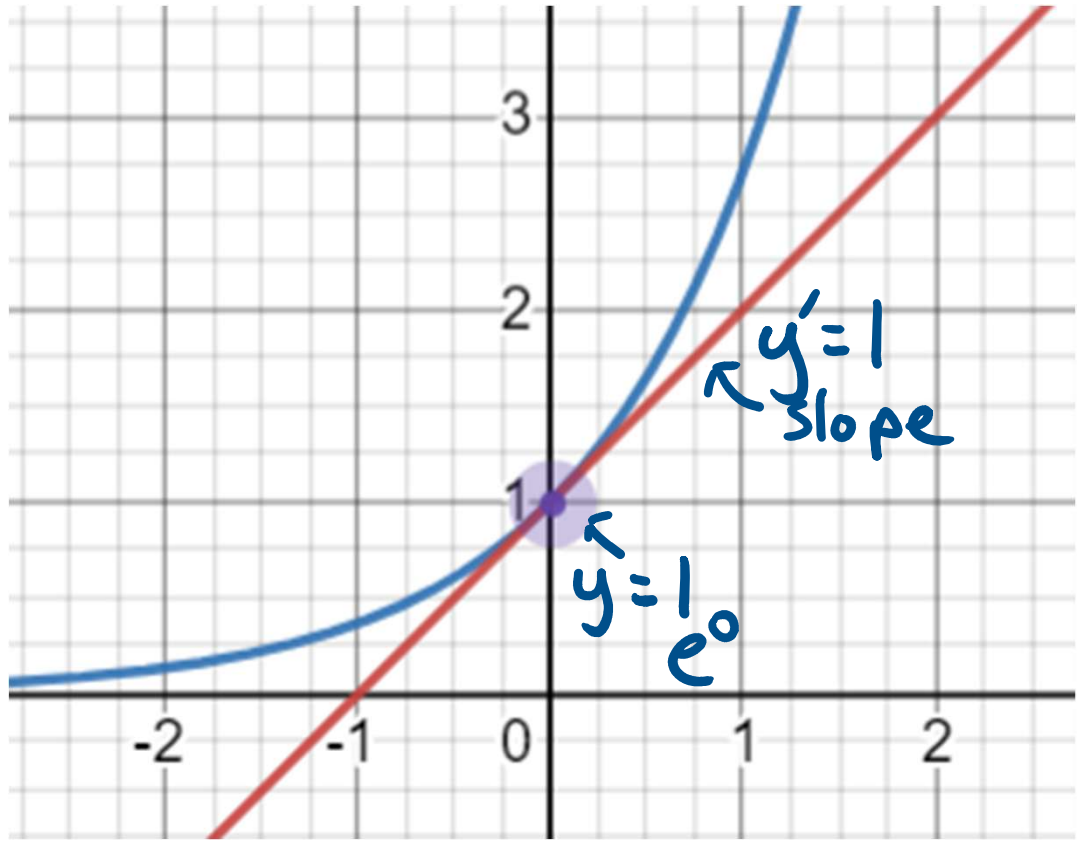
3

$a = 0$

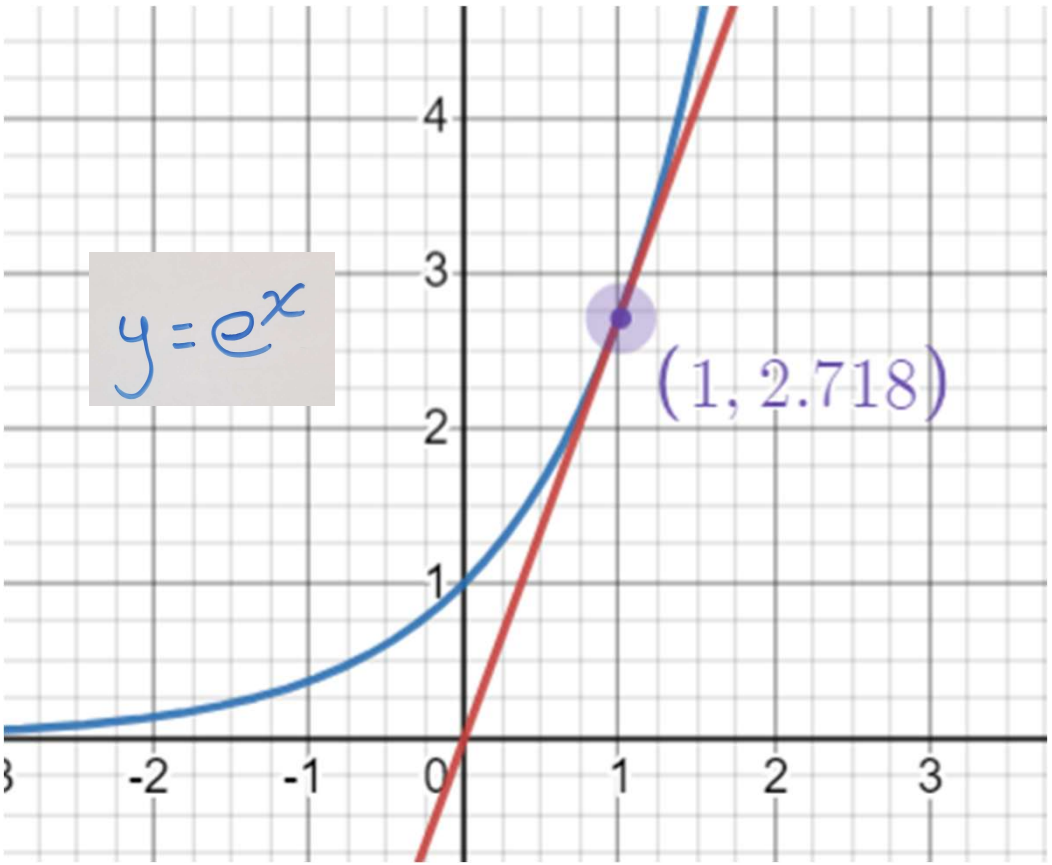
$___ \leq a \leq ___$ Step: $___$

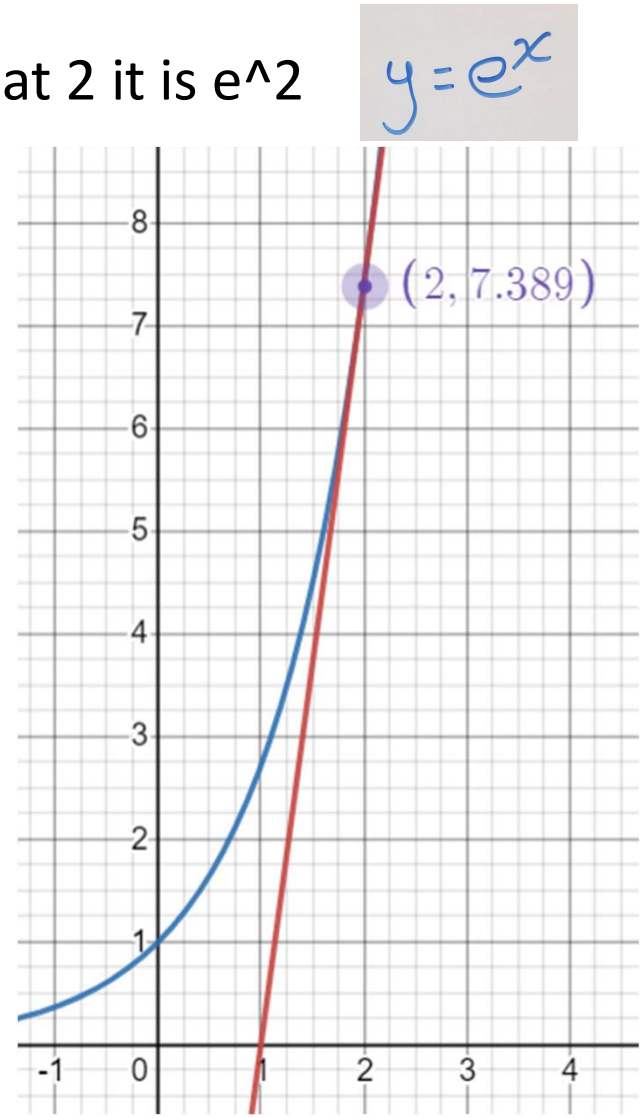
4

$y = g(a)(x - a) + f(a)$



When x is 1, the slope of the tangent is e and at 2 it is e^2



# Transformations of exponential functions

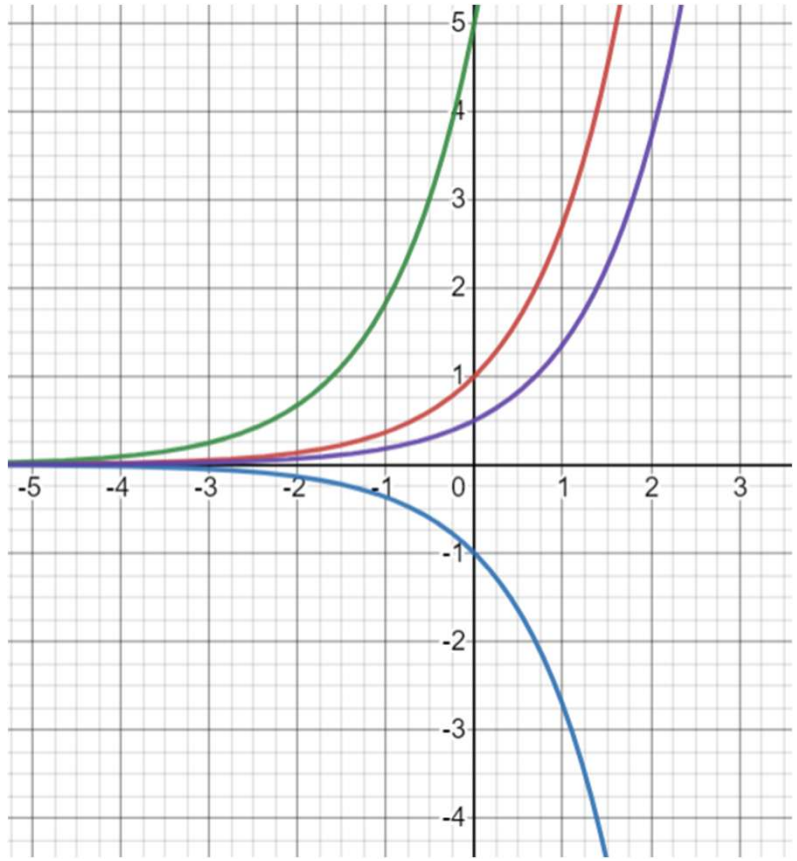
[Graph exponential functions using transformations | College Algebra (lumenlearning.com)]

$y = e^x$

$y = -e^x$

$y = 5e^x$

$y = \frac{1}{2}e^x$




$y = ne^x$

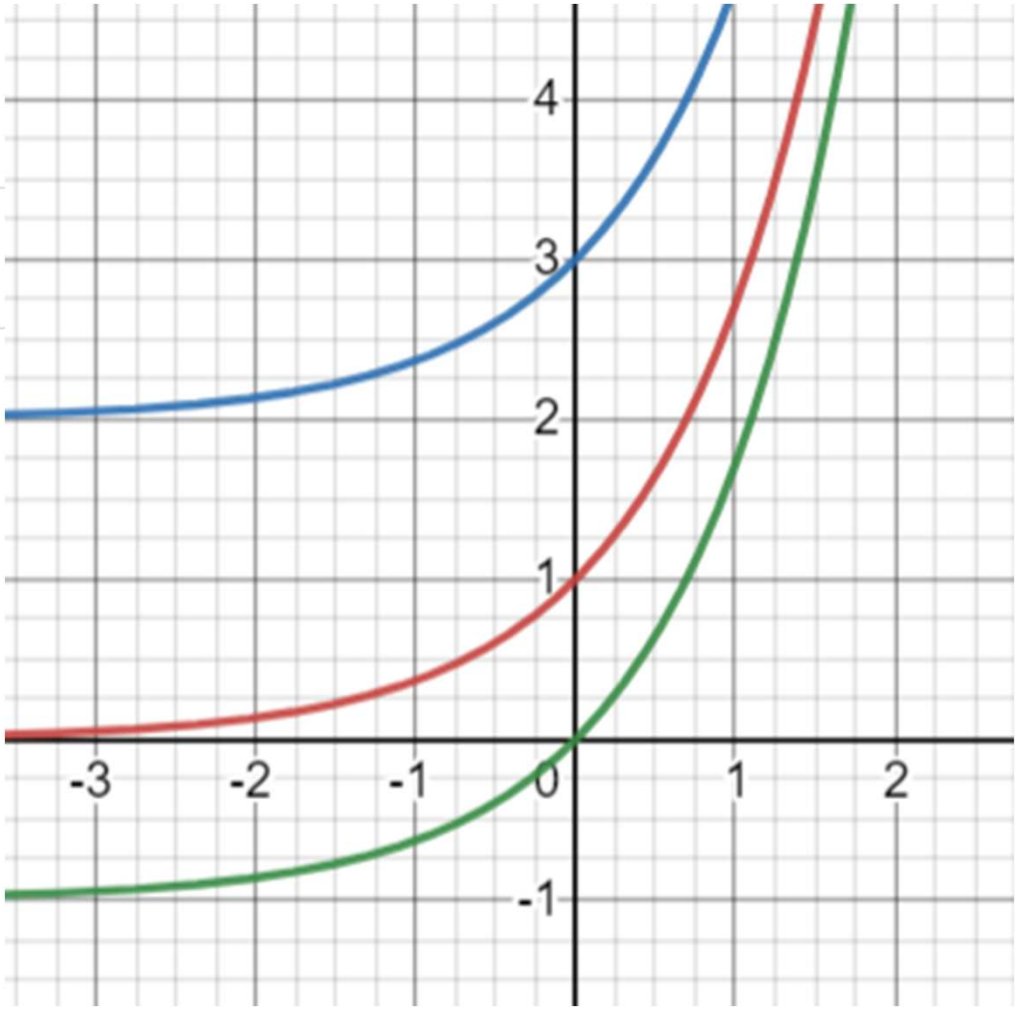
when $x = 0$ $ne^0$

yint: $n$

1 $y = e^x$

2 $y = e^x + 2$

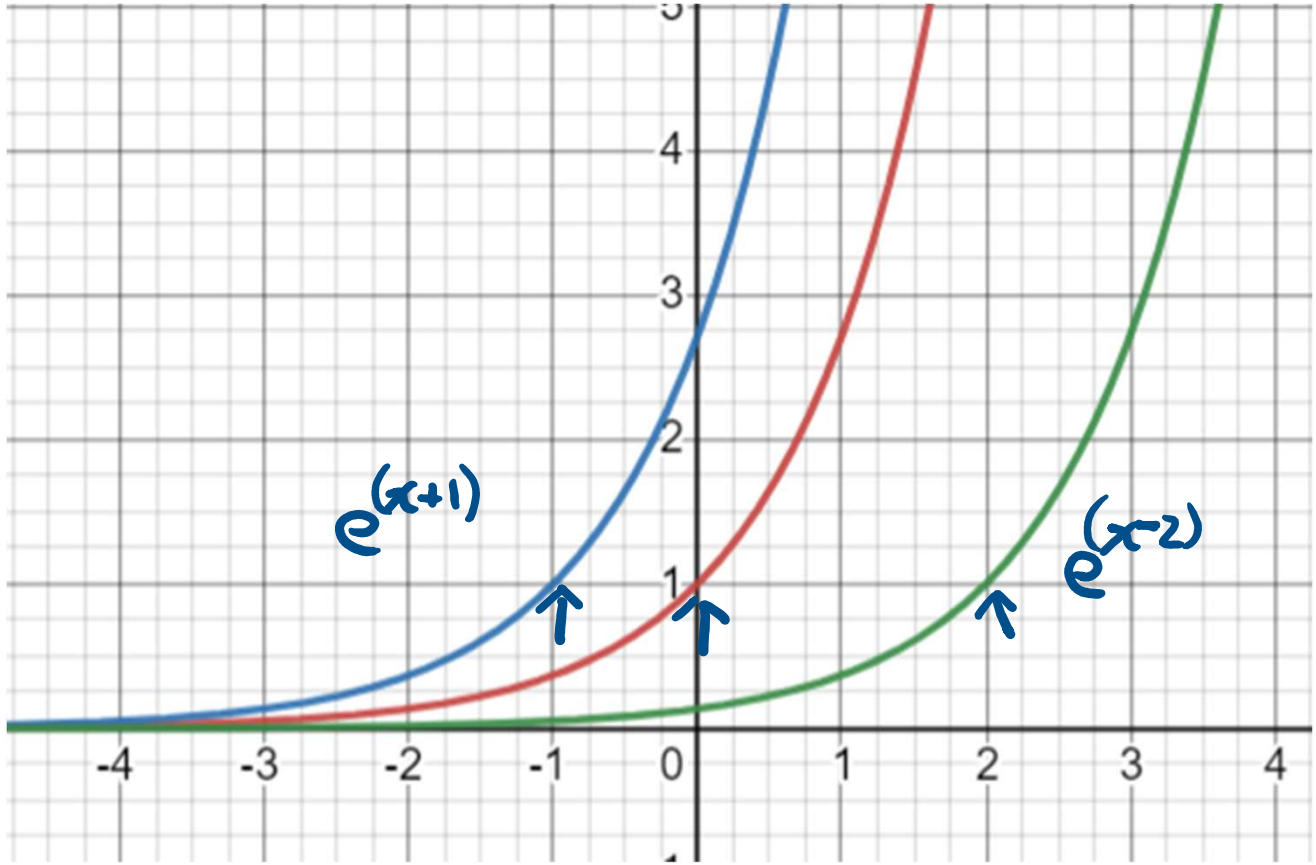
3 $y = e^x - 1$



1 $y = e^{x}$

2 $y = e^{(x+1)}$

3 $y = e^{(x-2)}$

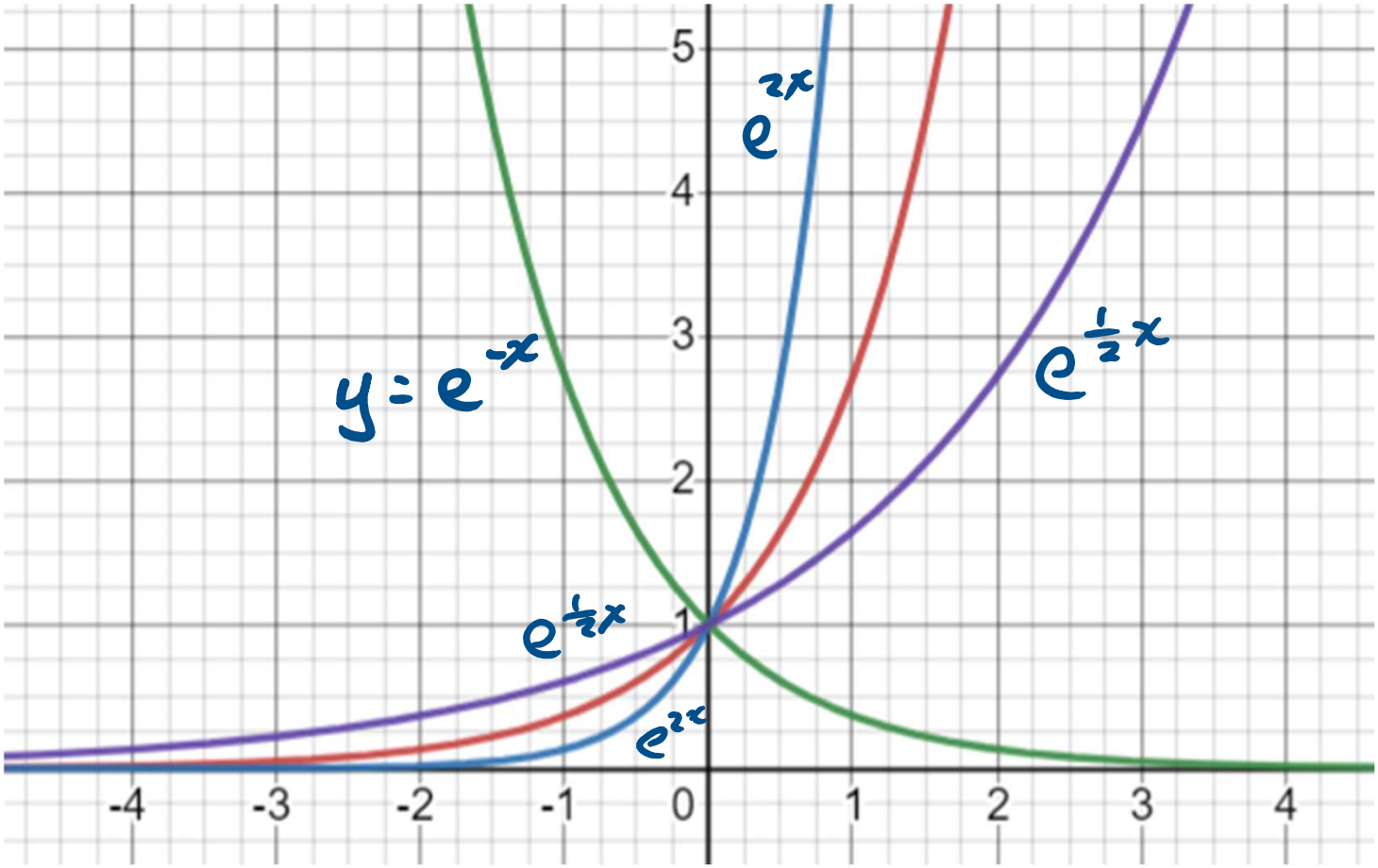


1 $y = e^{x}$

2 $y = e^{2x}$

3 $y = e^{(-x)}$

4 $y = e^{\frac{1}{2}x}$



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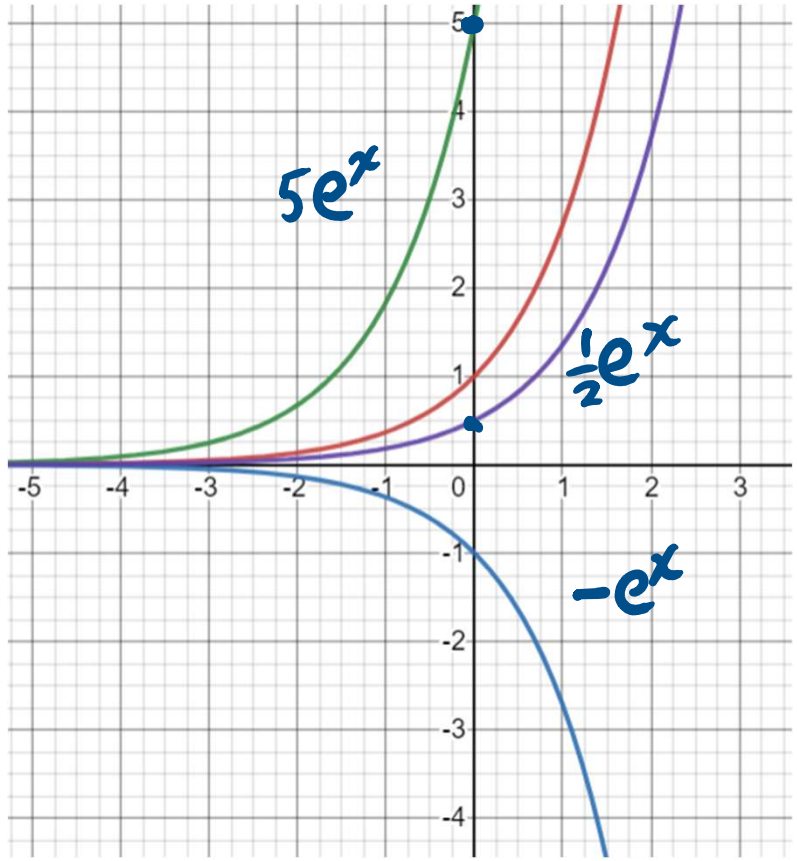

$y = e^{(-x)}$

$y = e^{x}$

2 $y = -e^{x}$

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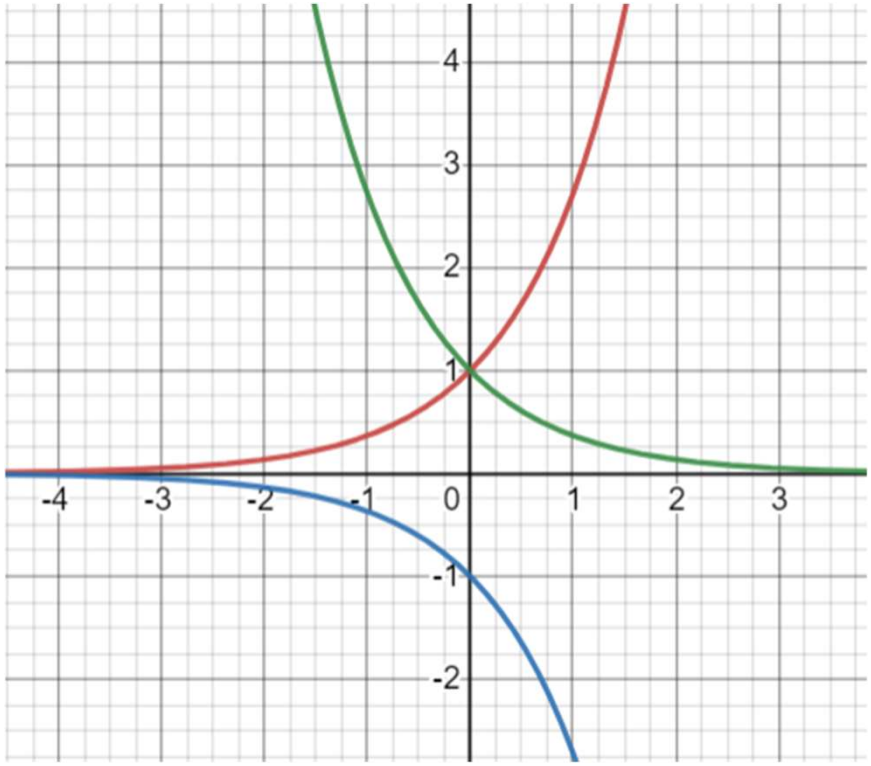


$y = -e^{x}$